

# Official Engineering Notebook

| Title: PLTW D&D | Date: NA/NA/NA | Revision: 0 |
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Description

Cover Page

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

The point after personal pitch presentations, we began the project for the class of PLTW Design and Development in Barrington High School. After dividing up into teams, we came to our groups with the following names: Arjun, Kushal, and Varun.

We came up with a decision matrix in order to decide which problem we could do within the span of a school year. After a project proposal, we narrowed our problems to three issues: Cable Management, Tennis ball machines, and 3D printing waste. We transferred research into summaries, providing sources, descriptions, and why this source is so important, and why they're credible.

With research commencing, the next steps will hopefully be soon.

**Cable Managment**

**Tennis Ball Machines**

**Decision Matrix**

**Cable Managment**

Problem (statement) Decision Matrix

| Problem Matrix (Arjun Shetty) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Priorities | Cost | Build Easibility | Circuitry Easibility | Ability to Test | Intrest | Time test | CAD Easibility | Coding Easibility | Have an idea already | | Final Weight |
| Weight | 1.50 | 1.50 | 1.00 | 1.00 | 1.50 | 1.00 | 1.50 | 1.00 | 2.00 | | |
| Cable Mangment | 8 | 8 | 9 | 10 | 5 | 9 | 7 | 10 | 1 | | 82 |
| Glasses & Contacts | 4 | 2 | 10 | 1 | 3 | 3 | 2 | 10 | 0 | | 40.5 |
| Tennis Ball Machines | 5 | 6 | 9 | 1 | 7 | 1 | 9 | 6 | 1 | | 59.5 |
| 3D Printing Waste | 3 | 7 | 2 | 4 | 10 | 7 | 9 | 7 | 1 | | 65.5 |
| Overheated Pillows | 6 | 8 | 10 | 1 | 3 | 1 | 8 | 10 | 0 | | 59.5 |
| Controller Stick Drift | 5 | 4 | 6 | 9 | 6 | 8 | 2 | 5 | 0 | | 53.5 |
| Note: The higher, the better. | 10: Cheap | 10: Easy to build | 10: No circuitry | 10: Easy to test | 10: Interesting | 10: Fast test | 10: Easy to CAD | 10: No code required | 1: True | | |
| | 1: Expensive | 1: Hard to build | 1: A lot of circuitry | 1: Hard to test | 1: Not interesting | 1: Slow test | 1: Hard to CAD | 1: Hard to code | 0: False | | |

Title: PLTW D&D

Date: 10/02/2026

Revision: 1

Description

An update on reserach summaries

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Today, after doing more research and refining our ideas, we met with Mr. Ytzen to review our remaining project concepts. He gave us the following feedback:**

- **Tennis ball machine and 3D printing waste: Both are real problems, but they may be too complex to solve fully as currently scoped. They need more refinement.**
- **Cable management: Not engaging enough to pursue, so we dropped it.**

**After the meeting, we focused on narrowing the two remaining concepts so each one addresses a single, manageable part of the problem. Arjun proposed one way to do this for 3D printing waste: grind failed or discarded prints into material that could be reformed and reused. We have not evaluated this idea yet. The period ended before we could discuss it, so we will pick it up at our next meeting.**

| Title: PLTW D&D | Date: 10/05/2026 | Revision: 2 |
|---|---|---|

Description

Options were narrowed by Mr. Ytzen's standpoint

Authors

Arjun Shetty
Kushal Chavda
Varun Nair

**We decided to focus more on recycling 3D printing waste since it was the problem we were most interested in pursuing. Cable managment was said to be the least entertaining (especially by Mr. Ytzen) while Tennis ball launchers and recycling 3d printing waste was proved to be difficult to build.**

**We also updated the layout and color scheme of our Canva Notebook by brightning the lines of the grid. This will allow the notebook to be easier to read, not to mention more appealing.**

**Finally, because our APA citations were still incomplete, we used a citation tool that Arjun recommended to finish them efficiently. This allowed our research to be more supported.**

| Title: PLTW D&D | Date: 10/06/2026 | Revision: 3 |
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Description

Narrowing down to one problem

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Varun began vibe coding the website to have some of the spesifications that are requred for the prodeuct website. The problem statements are listed on the home page, as well as progress over the other pages being developed. While we have Element A in progress, the personal profile section has yet to be started.**

**With the website in play, we can now start working our way on the progress we have by adding in reporst, sources, and a lot more to get the project going.**

Title: PLTW D&D

Date: 10/07/2026

Revision: 4

Description

Website progress

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**The website is soon to have the personal profile page. Arjun submitted information to Varun for him to intergrade it into the website, that way the process can commence, while Kushal completed some more research about 3d printing waste, focusing on pollution.**

**Arjun also managed to pull PDF from the sources to be stored as resources, that way we will have them for use to prove our problem as valid.**

| Title: PLTW D&D | Date: 10/08/2026 | Revision: 5 |
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Description

Continued progress

Authors

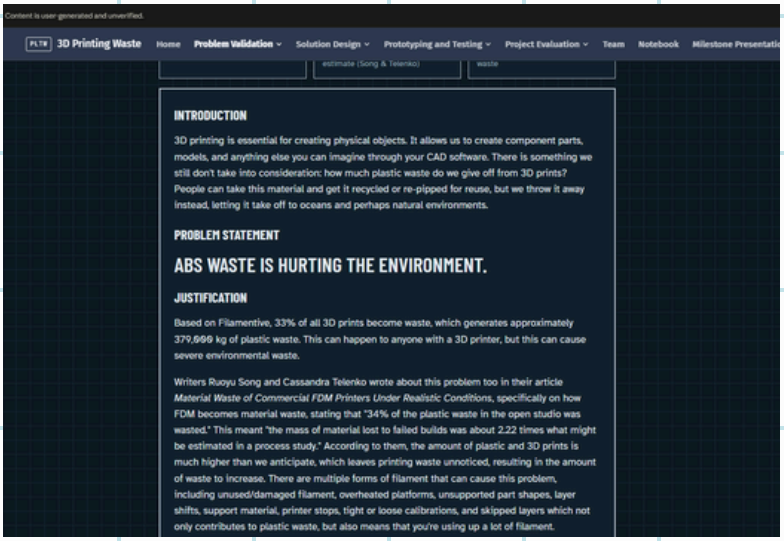
Arjun Shetty

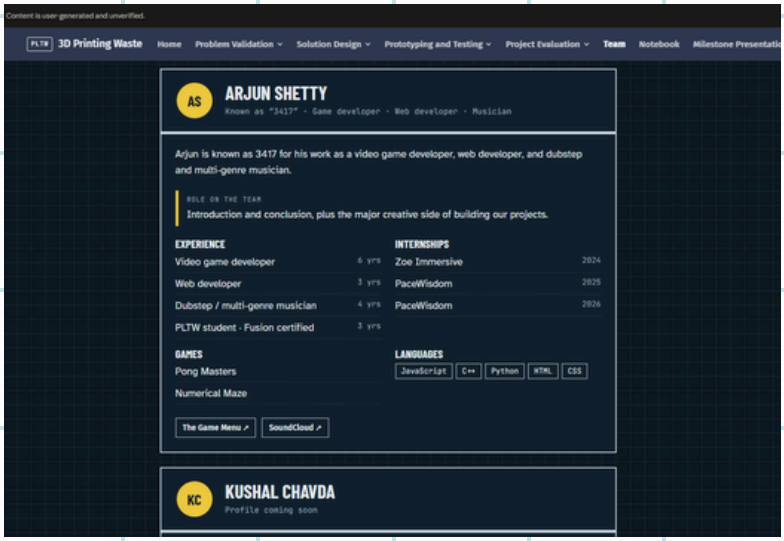
Kushal Chavda

Varun Nair

**Varun finally got the Team page initiated, starting with Arjun's background. On top of that, Element A–written by Arjun (and Kushal), vibe-coded by Varun, and sourced by everyone–has been complete and put on the site. It may be up for review based off a few possible mistakes by Claude, but other than that, major progress was FINALLY made! The next steps are to get an image (or two) on the website, and to find out what Element B is.**

## Website




| Title: PLTW D&D | Date: 10/09/2026 | Revision: 6 |
|---|---|---|

Description

Major progress on the Team page and Element A

Authors

Arjun Shetty
Kushal Chavda
Varun Nair

**Arjun and Kushal began working on patent and product research (Element B). We found a lot of products that were in interest to us; however, they had many flaws including price and manual labor. Varun was absent, though he previously said he had done some research about patents which will be useful once he comes back.**

**Arjun also managed to amplify Element A with some formating, which will soon be implimented by Varun. This was done to prevent Claude from taking away the creative aspect of his writing, and to ensure that no false statement was put in.**

**Progress has been made today, and we hope to keep doing that.**

| Title: PLTW D&D | Date: 10/12/2026 | Revision: 7 |
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Description

Beginning Element B and Revising Element A

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Varun ran out of credits to update the website, which meant that we couldn’t do anything about the website. Due do a reduced amount of assistance, we only managed to get a single patent searched, and considered another in chinese.**

| Title: PLTW D&D | Date: 10/13/2026 | Revision: 8 |
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Description

Element B

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**After some more research on Element B, it was time to make progress on element A. In order to complete element A, we needed some individuals to take a poll in order to get a rundown on how much waste is actually being wasted, and how much money was being spent on filament. This was posted in Reddit, and now all we have to do is wait.**

| Title: PLTW D&D | Date: 10/14/2026 | Revision: 9 |
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Description

The Poll

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**We were stuck at 19 responses with our form, making data useless this day. Our goal of 200 will be hard to achive from here, so it's crutial that we get a move on with these responses, this way we would have a variaty of responses.**

**During class, Kushal worked on Element B to fix a mistake our group had made. We accidentally left out one of our product research, however it was a simple fix.**

| Title: PLTW D&D | Date: 10/19/2026 | Revision: 10 |
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Description

Responses + Revisions

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Today, our group focused mainly on Element A since we would get feedback if we submitted before the morning of 10/21/2026 . Kushal and Varun worked on getting the survey more widespread and looked at the Feedback from the Reddit Post. Some users said the questions were not clarified enough so we had to remove one.**

**Arjun also managed to create a graph off Google Sheets that would update live. Unfortunatly this graph is only avalible to those within the Barrington220 community. Once we're ready, we can add the graph into our website using an HTML iframe element.**

| Title: PLTW D&D | Date: 10/20/2026 | Revision: 11 |
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Description

Element A work

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Element B has been drafted and posted on the website in a standard form. With Varun's Claude prompt, our website not only has the product and patent research, but also has buttons that take you to the original source, which saves us some time. We have yet to submit Element A for the last time, so our hope is to get that done before the deadline.**

Title: PLTW D&D

Date: 10/22/2026

Revision: 12

Description

Element B draft and Element A plans.

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Element A has been submitted for final grading, and a draft of Element B is on the line. We will be here waiting until the next week**

| Title: PLTW D&D | Date: 10/23/2026 | Revision: 13 |
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Description

Submissions

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Today, we got good progress on Element A. We needed an expert on our project who is interested in our problem so we called companies who had 3d printing services. One of them was the Barrington Library. Kushal and Varun called them, asking about their opinions of 3d printing waste. They said they would call us back. At around 3:30, they called Kushal back and they said they were interested. When asked about his opinions, he replied with, "I wish to recycle them". He also said a recycler would be really beneficial at the Library. This is currently a really good expert to consider. Kushal also ended up getting his name which was Mike.**

**A few hours later, Arjun managed to pull the HTML to add the Google Sheets Graph to Element A, as well as a major writing update to form a better and more appealing report. Next, Element B will recive images in order to succeed.**

| Title: PLTW D&D | Date: 10/26/2026 | Revision: 14 |
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Description

Extreme major progress.

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Element C has been started, as our introduction and stakeholders have been placed. We came up with 6 requirements thus far, and plan to enhance them even more as the days progress.**

**Meanwhile, Arjun experimented with the new website by using his own repository. The results: it needs to take bug fixing. The confusing thing was that everything was fine during testing, and by the time it was uploaded, images were missing and some elements didn't seem to work. This will be polished tommorrow or Thursday.**

| Title: PLTW D&D | Date: 10/27/2026 | Revision: 15 |
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Description

Element C and potential bug fixing.

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**There isn’t much that happened. All that happened this day follows:**

**Arjun managed to build and intigrate parts of Element A, B, and C into HTML, while Varun did it with Claude. While Varun’s edition is being hosted, Arjun’s is also hosted, and our team will see which way is more efficient.**

**During the end, a new requirement was added: Saftey, which would be vital in the long run, as our build may need heating and sharp elements.**

| Title: PLTW D&D | Date: 10/28/2026 | Revision: 16 |
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Description

The little that happened

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**There isn’t much that happened. All that happened this day follows:**

**Arjun got all parts of Element C into place for his edition of the website, while varun has a placeholder for his interview notes. When these parts are done, Elements B and C will be submitted.**

**The Project proposal presentation has also been started, and once all elements are complete, we will make our way to pushing this proposal over the finish line.**

| Title: PLTW D&D | Date: 10/29/2026 | Revision: 17 |
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Description

The little that happened (again)

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**The presentation has been started, and has got a few startup slides to help us get some information down that will help support our product. We are still waiting on expert opinions, which may become a doubtful source at this point on.**

**Element B was subbmitted, and Element C's draft was too, which gives us an upper hand during our break.**

**Other than updating the website navigation menu and a few other cool features, this was all that happened.**




| Title: PLTW D&D | Date: 10/30/2026 | Revision: 18 |
|---|---|---|

Description

The beginning of the proposal!

Authors

Arjun Shetty
Kushal Chavda
Varun Nair

**The presentation has taken major progress, and is now at 31 slides, where some purpose themselves as transitions to avoid overload for viewers. With most of our information in place, we just need our expert opinions, which we are still pending on.**

**Other than that, we submitted element C, and our website's justification segment may come to a close.**

| Title: PLTW D&D | Date: 11/1/2026 | Revision: 19 |
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Description

Major progress

Authors

Arjun Shetty

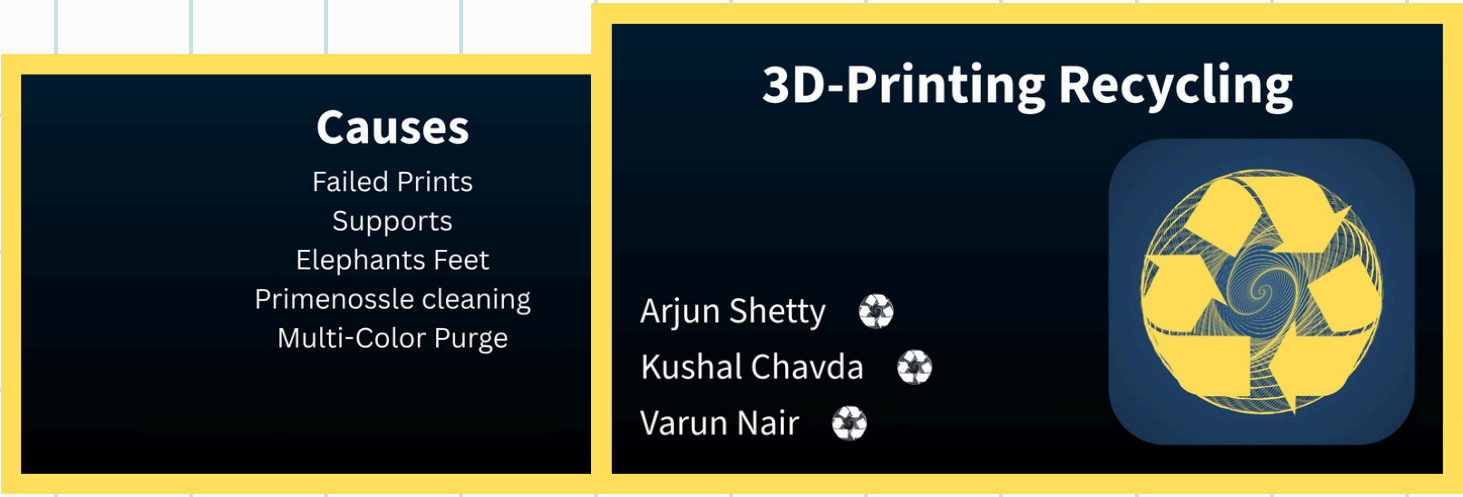
Kushal Chavda

Varun Nair

**The presentation has been submitted without the expert views, but there's still hope. As long as we can get these views before 8th period tommorrow, we can have a finished presentation.**

**These views are important, as it provides an insight on what we're really dealing with, so to have this in our presentation will give us an upper hand.**

**At the library on Tuesday (11/3/26), our expert (Mike) walked us through their workspace and explained which machines they used. They had three 3D printers and one really big one. He states they used PLA filament which is a biodegradable material and perfect for recycling. Mike also indicated that a product that recycles filament would be of the best interest for the Barrington Area Library and possibly other locations.**




| Title: PLTW D&D | Date: 11/3/2026 | Revision: 20 |
|---|---|---|

Description
Submission and Expert Interview

Authors
Arjun Shetty
Kushal Chavda
Varun Nair

We have our expert opinions on 3D-printing recycling in our presentation. Now it's time to start writing some notes. Our plan is to decide which part will be said by who, and in order to do that, we would use the notes feature in Canva. There is a chance Varun will be absent, so his segment may be divided between Kushal and Arjun.

## Michael Campagna (Mike)

- Biodegradable Filament Polylactic Acid (PLA)
- PLA is only commercial composted.
- Strong interest in recycling filament.







| Title: PLTW D&D | Date: 11/4/2026 | Revision: 21 |
|---|---|---|

Description
Expert Opinions

Authors
Arjun Shetty
Kushal Chavda
Varun Nair

**We continued working on the presentation after adding a few slides. Kushal began writing the notes for the products and patients, while Arjun began writing them for the causes and the product requirements. We decided it was best to split up the roles based on who's more knowledgeable about that section. We plan to present on Monday 11/9/2026 and we have to keep our presentation at around 8 minute.**

| Title: PLTW D&D | Date: 11/5/2026 | Revision: 22 |
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Description

Slide Progress

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Our presentation is complete, and our roles have been assigned. Our presentation day has been set to Monday, and after writing some notes down on what we're gonna say, we should have no problem executing the presentation.**

**Meanwhile, Arjun began implimenting the expert views in the website before it was too late. He will be waiting for Kushal's answer, but if there isn't one, this segment will have to be freeballed based off the information in the slide deck.**

| Title: PLTW D&D | Date: 11/6/2026 | Revision: 23 |
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Description
Slide Completion

Authors
Arjun Shetty
Kushal Chavda
Varun Nair

**Arjun implimented the expert views section. Because there was not written summary of what had happened, he ended up freeballing it. Tommrorrow, he will check in with Kushal to ensure that the writing is up to standard.**

Title: PLTW D&D

Date: 11/8/2026

Revision: 24

Description

Element A update

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**The presentation has been given, and was thrown by someone. Though the rest of us gave it our all, some crucial elements were not provided correctly because of someone's lack of effort. We MUST take this project seriously, as our grade depend on it.**

Title: PLTW D&D

Date: 11/9/2026

Revision: 25

Description

Presentation day

Authors

Arjun Shetty

Kushal Chavda

Varun Nair

**Our group had a talk with Mr. Ytszen and he suggested that we should have a change in our problem. We will start brainstorming new problems and will have to redo all of the work we have done.**

**This means all our research and our designs will be scrapped into the dumpster, and so we start over once again...**

| Title: PLTW D&D | Date: 11/10/2026 | Revision: 26 |
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Description
Major Letdown

Authors
Arjun Shetty
Kushal Chavda
Varun Nair